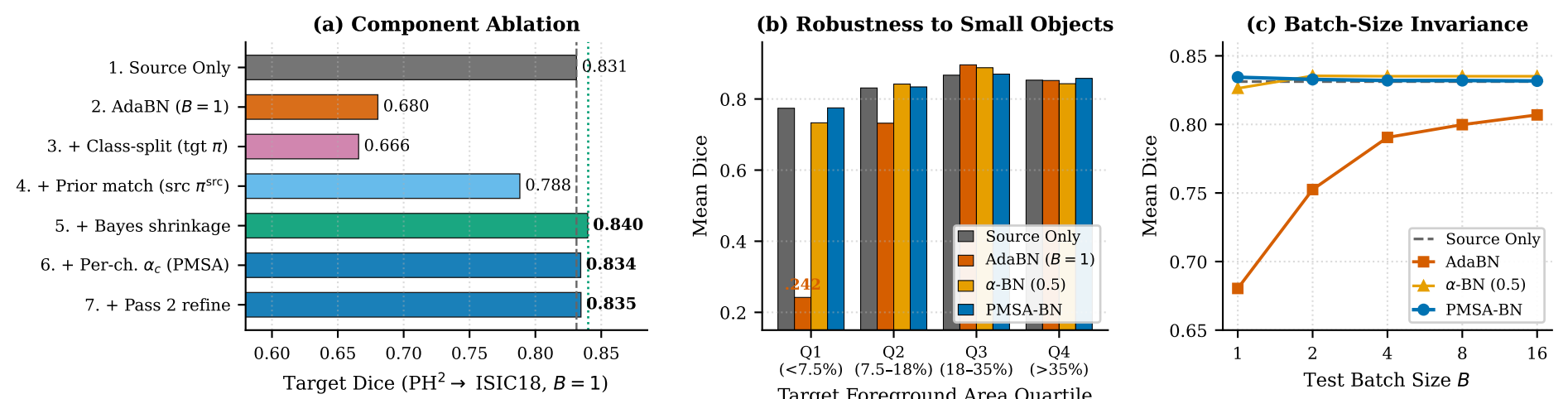
(a) Component Ablation
1. Source Only
2. AdaBN ($B = 1$)
3. + Class-split (tgt π)
4. + Prior match (src π^{src})
5. + Bayes shrinkage
6. + Per-ch. α_c (PMSA)
7. + Pass 2 refine
0.831
0.680
0.666
0.788
0.840
0.834
0.835
0.60
0.65
0.70
0.75
0.80
0.85
Target Dice (PH^2 → ISIC18, $B = 1$)
(b) Robustness to Small Objects
Mean Dice
0.2
0.4
0.6
0.8
.242
Source Only
AdaBN ($B = 1$)
α-BN (0.5)
PMSA-BN
Q1 (<7.5%)
Q2 (7.5–18%)
Q3 (18–35%)
Q4 (>35%)
Target Foreground Area Quartile
(c) Batch-Size Invariance
Mean Dice
0.65
0.70
0.75
0.80
0.85
Source Only
AdaBN
α-BN (0.5)
PMSA-BN
1
2
4
8
16
Test Batch Size B